the value chain

Innovate to Accelerate

SAP API Management

- Understand what API management is all about
- API Portal and Developer Portal
- Some useful links
- Questions?

- Create simple digital experiences for your consumers, partners, and employees.
- Publish, promote, and oversee APIs in a secure and scalable environment.
- Cloud Foundry and Neo gives you the ability to run SAP API Management

Components of SAP API Management

API Management

API Runtime

API Portal

API Business Hub Enterprise

API Analytics

API Designer

Deploy and productively use your APIs. Applications consume the API Runtime, request for API authentication, and access.

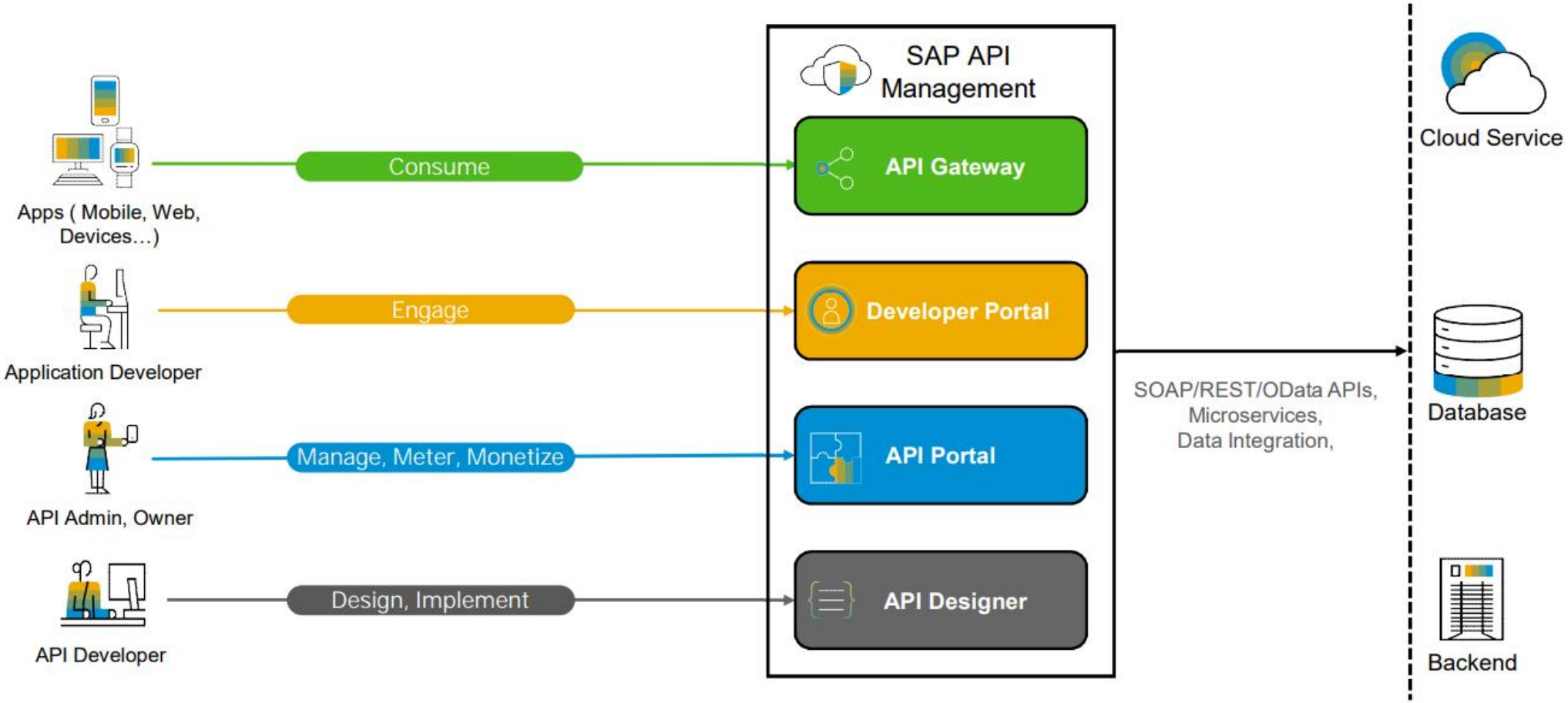
The one-stop-shop to create, secure, and publish API Proxies. This is the place for easy discovery of APIs, and the API Administrator, can manage, meter, secure APIs, as well as define and publish rate plans

Self-service for application developers to discover, browse, and explore APIs, subscribe to rate plans, and build apps

Provides powerful analytical tools to track your API usage. Use API Analytics to collect information

API developers can define, implement, and document APIs. It provides open API support, and a variety of outputs can be generated.

High-level Architecture



API Management Features

API Portal

API Designer

Proxies

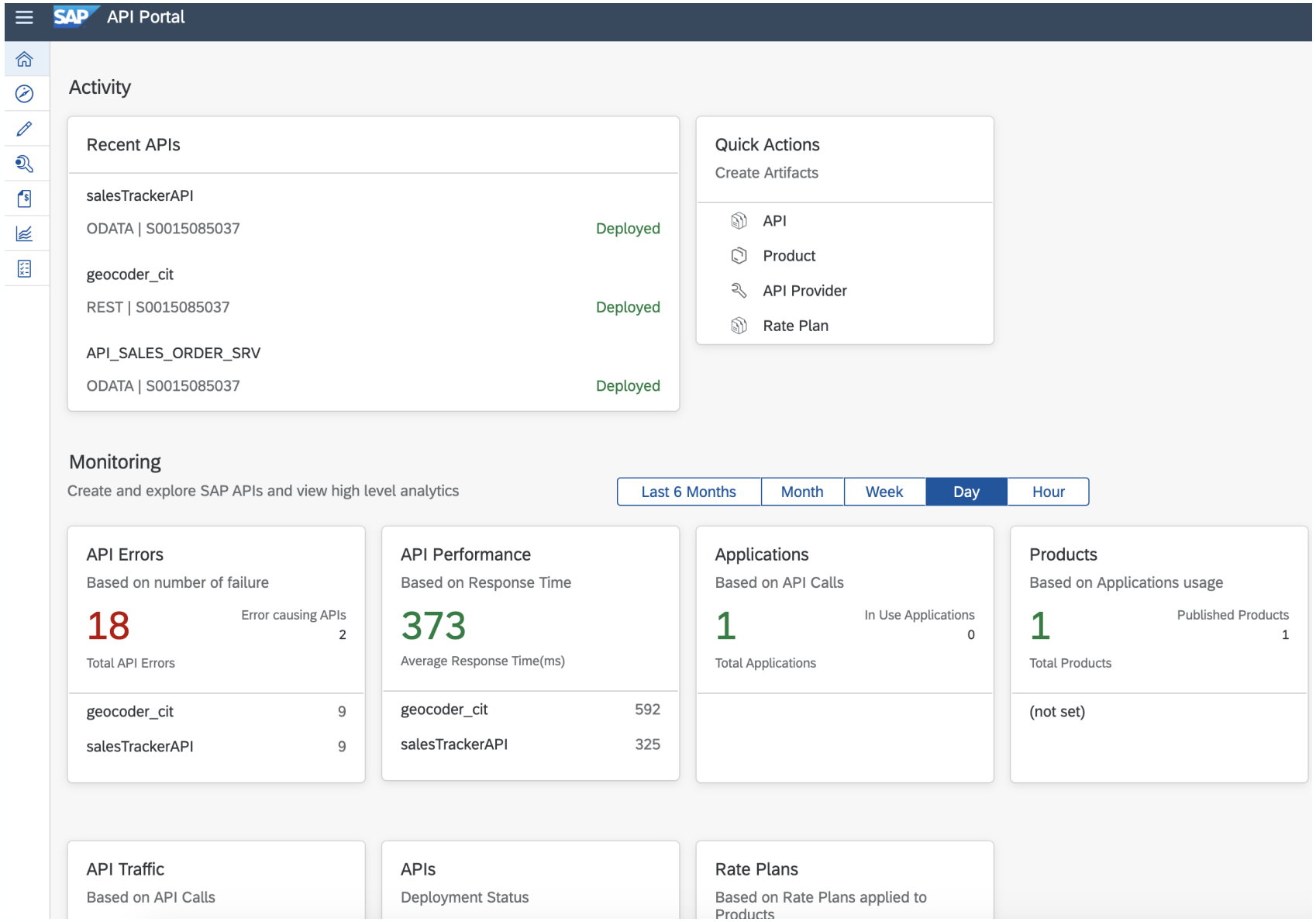
Configure runtime of API

Re-use configurations through templates

Security

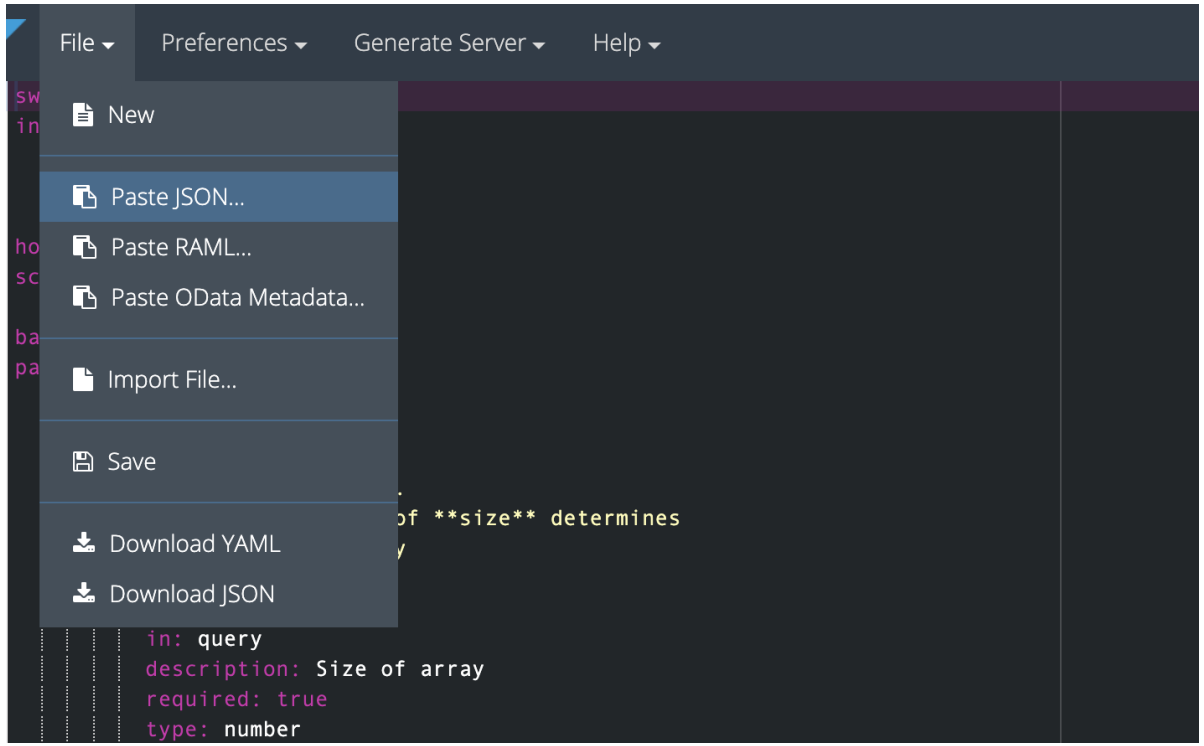
Analytics

API Portal


API Portal
Activity
Recent APIs
salesTrackerAPI
ODATA | S0015085037
Deployed
geocoder_cit
REST | S0015085037
Deployed
API_SALES_ORDER_SRV
ODATA | S0015085037
Deployed
Quick Actions
Create Artifacts
API
Product
API Provider
Rate Plan
Monitoring
Create and explore SAP APIs and view high level analytics
Last 6 Months
Month
Week
Day
Hour
API Errors
Based on number of failure
18
Total API Errors
Error causing APIs
2
geocoder_cit 9
salesTrackerAPI 9
API Performance
Based on Response Time
373
Average Response Time(ms)
geocoder_cit 592
salesTrackerAPI 325
Applications
Based on API Calls
1
Total Applications
In Use Applications
0
Products
Based on Applications usage
1
Total Products
Published Products
1
(not set)
API Traffic
Based on API Calls
APIs
Deployment Status
Rate Plans
Based on Rate Plans applied to Products

API Portal: Designer

Design-first approach



OpenAPI support

Import your APIs

RAML
Odata
JSON

Generate Server Stubs

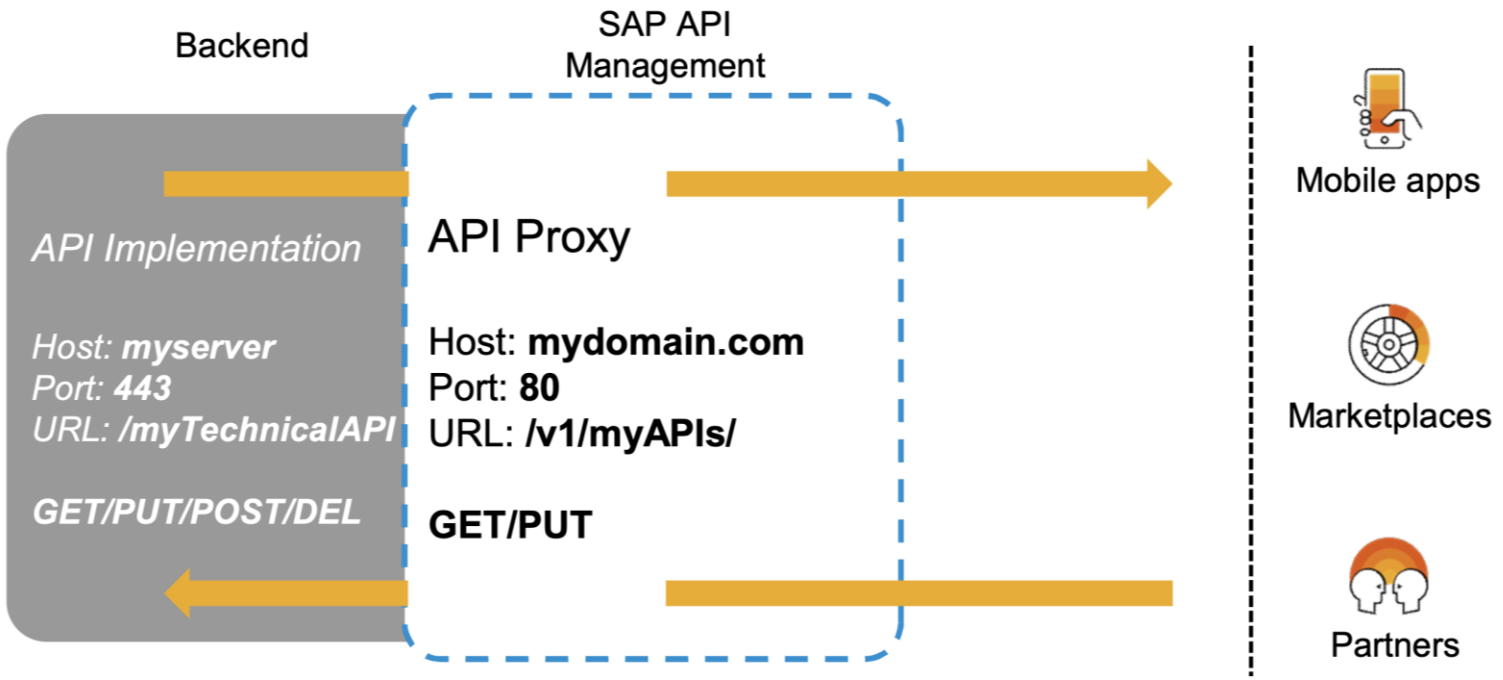
NodeJS
JAX-RS
Spring

Generate API in API Management

API Portal: Proxies

- Customers are not changing the actual API implementation
- Defining **API Proxies** to existing APIs
- **Hide internal information and add specific behavior**

- OpenAPI
- ODATA



Configure, don’t code

- Policy Editor - graphical

API Artifacts for salesTrackerAPI

Edit | Policy Template | Cancel

Policy Editor

Flows

ProxyEndpoint

PreFlow	8
PostFlow	0
BusinessPartn...	0
ContactSet	0
ProductSet	0
RegenerateAll...	0
SalesOrderLin...	0
SalesOrderSet	0

Created Policies

- CallGeocodingAPI
- defaultRaiseFaultP...
- readAddressDataJ...
- readAddressDataX...
- readGeoMapRespo...
- responsehandler

Scripts

- responsehandler.js

setCredentia...

setCORS | responsehand... | readGeoMapRe... | CallGeocodin... | readAddressD... | readAddressD... | setMetadata

setCredentials

Condition String

```
<!-- This policy can be used to create or modify the standard HTTP request and response messages -->
<AssignMessage async="false" continueOnError="false" enabled="true" xmlns='http://www.sap.com/apimgmt'>
    <Set>
        <Headers>
            <Header name="Authorization">Basic UzAwMTUwODUwMzc6SkBuQGRrMDkxNQ==</Header>
        </Headers>
    </Set>
    <IgnoreUnresolvedVariables>false</IgnoreUnresolvedVariables>
    <AssignTo createNew="false" type="request">request</AssignTo>
</AssignMessage>
```

- Invalidate Cache
- Lookup Cache
- Populate Cache
- Quota
- Reset Quota
- Response Cache
- Spike Arrest

Mediation Policies

- Access Entity
- Assign Message
- Extract Variables
- JSON to XML
- Key Value Map Operations
- Raise Fault
- XML to JSON
- XSL Transform

Extension Policies

- JavaScript
- Message Logging
- Message Validation

Policies

- Pre-defined policies, grouped into 4 categories

- Security
  - On every level: message level, network level, user level
- Traffic Management
  - Protect backend systems against high amount and concurring requests
- Mediation
  - Transform data that froms from or to your actual API implementation
- Extension
  - Create your own specific API runtime behavior by coding it

Security Policies

- Basic Authentication
- DecodeJWT
- GenerateJWT
- JSON Threat Protection
- OAuth v2.0
- OAuth v2.0 GET
- OAuth v2.0 SET
- Regular Expression Protection
- SAML Assertion Generation
- SAML Assertion Validation
- Verify API Key
- VerifyJWT
- XML Threat Protection

Traffic Management Policies

- Access Control
- Concurrent Rate Limit
- Invalidate Cache
- Lookup Cache
- Populate Cache
- Quota
- Reset Quota
- Response Cache
- Spike Arrest

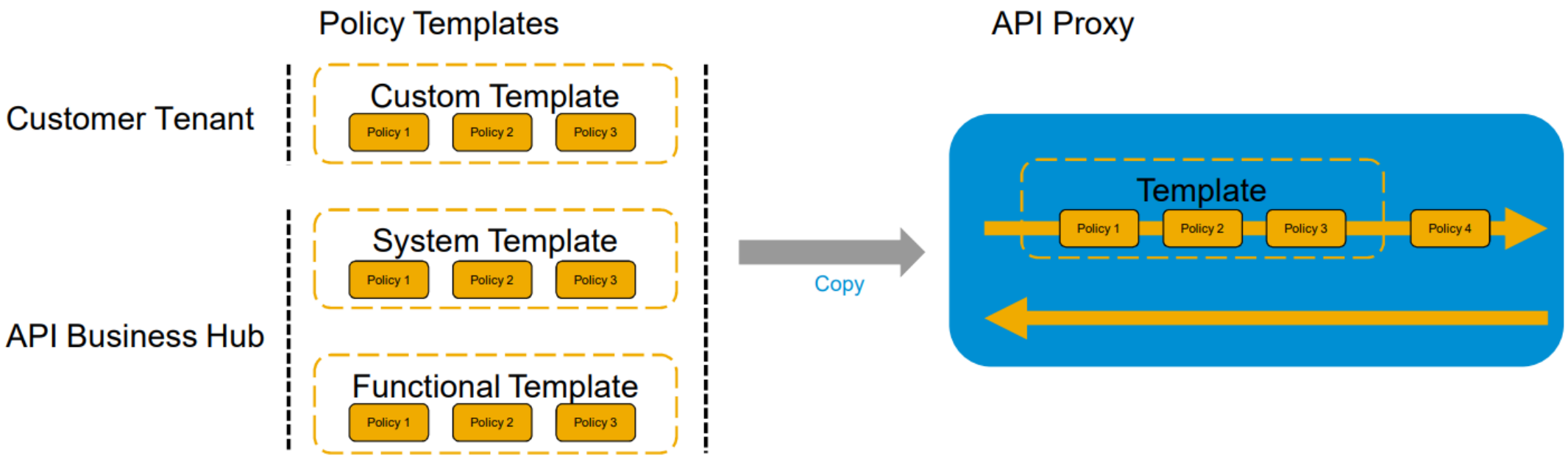
Mediation Policies

- Access Entity
- Assign Message
- Extract Variables
- JSON to XML
- Key Value Map Operations
- Raise Fault
- XML to JSON
- XSL Transform

Extension Policies

- JavaScript
- Message Logging
- Message Validation
- Open Connectors
- Python Script
- Service Callout
- Statistics Collector

Package and reuse existing policy sets



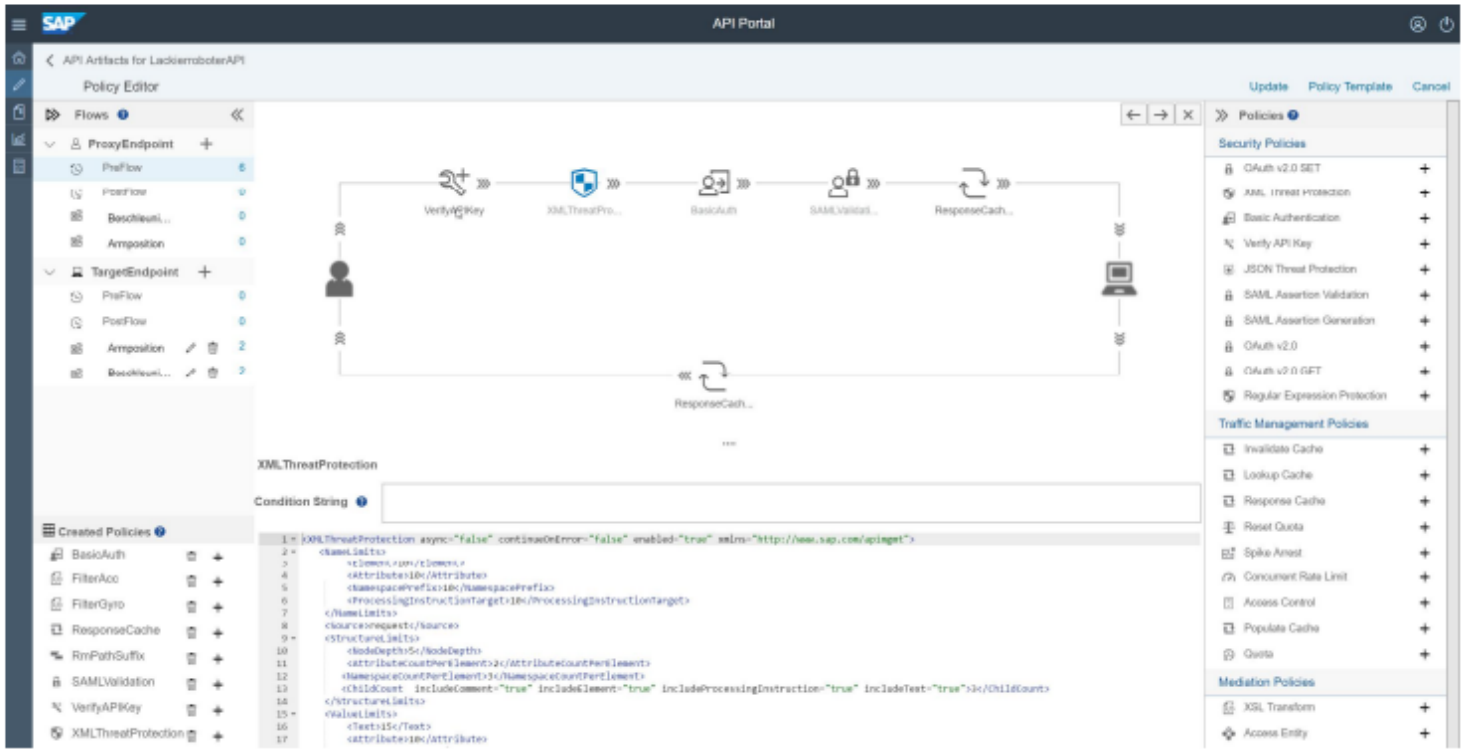
→ re-use configurations through templates

Security on All Levels

Threat protection

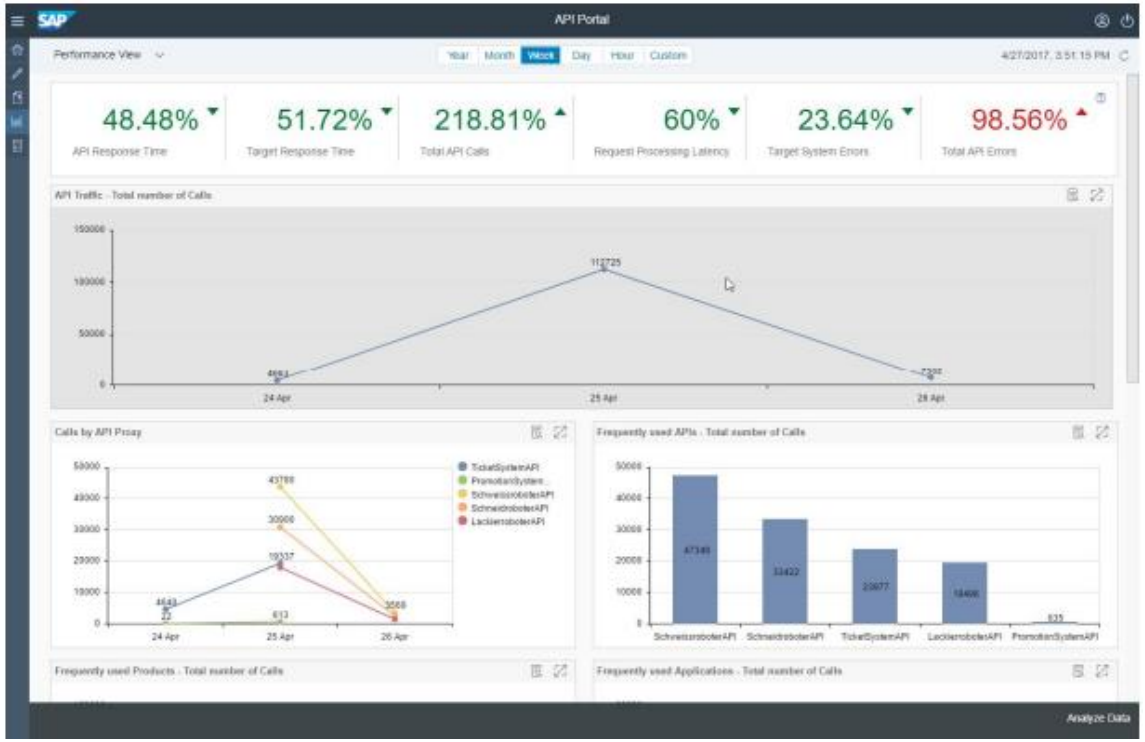
Authentication and authorization

Role-based access control



Analyzing APIs

- Technical Monitoring
  - API Administrator has access to pre-defined dashboards
  - Overview of technicall runtime information such as number of calls, backend response times, api proxy response time, errors..
  - Quickly identify issues & start remediation
  - Debug tool to trace the behavior of API Proxy at runtime
- Usage Analysis
  - “who uses in what API in context” ; “what are successfull applications”
  - Possible through usage of API Key, intagrated in API Management Analytics
- Custom monitoring
  - Analyse API Layer through specific dashboards
  - Create own dashboards with pre-defined filtering and metrics in an easy and graphical way
  - Drill-down capabilities to create interactive reports for purposes



Developer Portal

DISCOVER

CONSUME

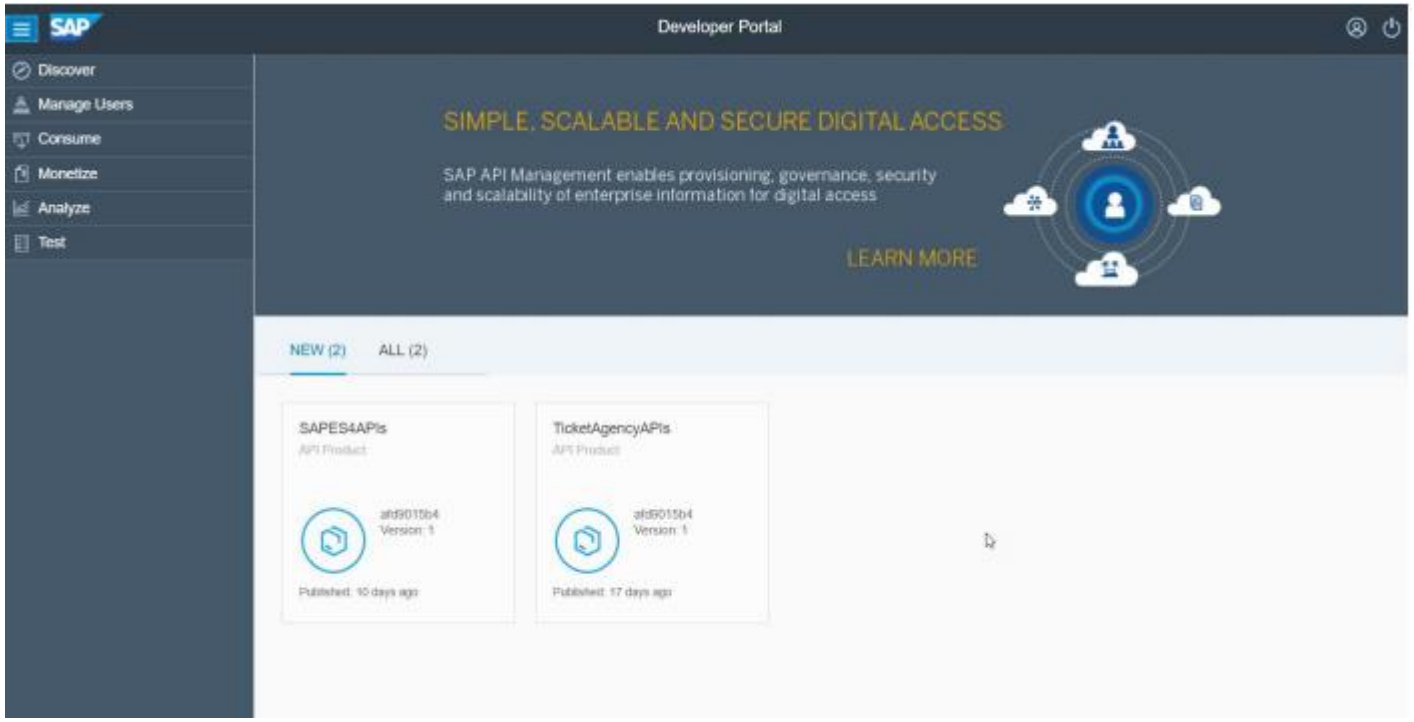
MONITOR

PUBLISH

ANALYTICS

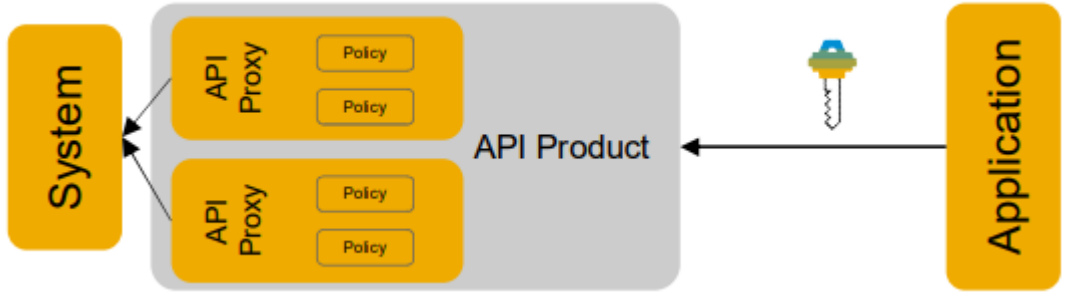
Developer Portal

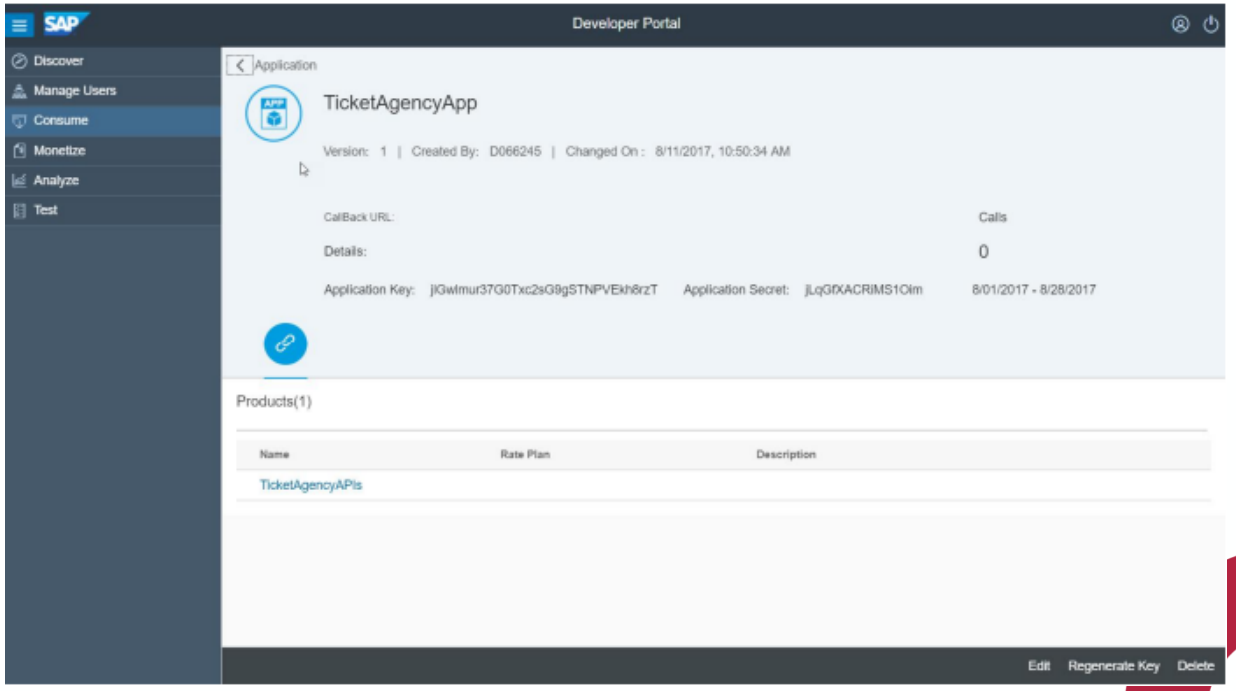
- main entry-point for the application developer
- Developers self-register and can:
  - Discover available products and the API Proxies which are included,
  - Test APIs online,
  - Subscribe to an API product by creating an Application (a represenation of the „real“ application which he will develop),
  - Manage his API keys.



Developer Portal – App Key

- Each Application is tied to a Developer through an API Key: the API key identifies the Application, and is used primarily for analytics
  - It is a best-practice to use API keys
  - API keys are managed by the Application Developers
  - API Key is not a security feature!





Documentation

- If metadata is available, documentation of an API is created in an automated way
  - OpenAPI specification can be read and imported into the API (& RAML)
  - Same happens when using SAP Gateway as API backend system or when using Odata REST APIs
- Can be refined or created from scratch, using the embedded editor of the API Portal

Title:
Sales Order (A2X)

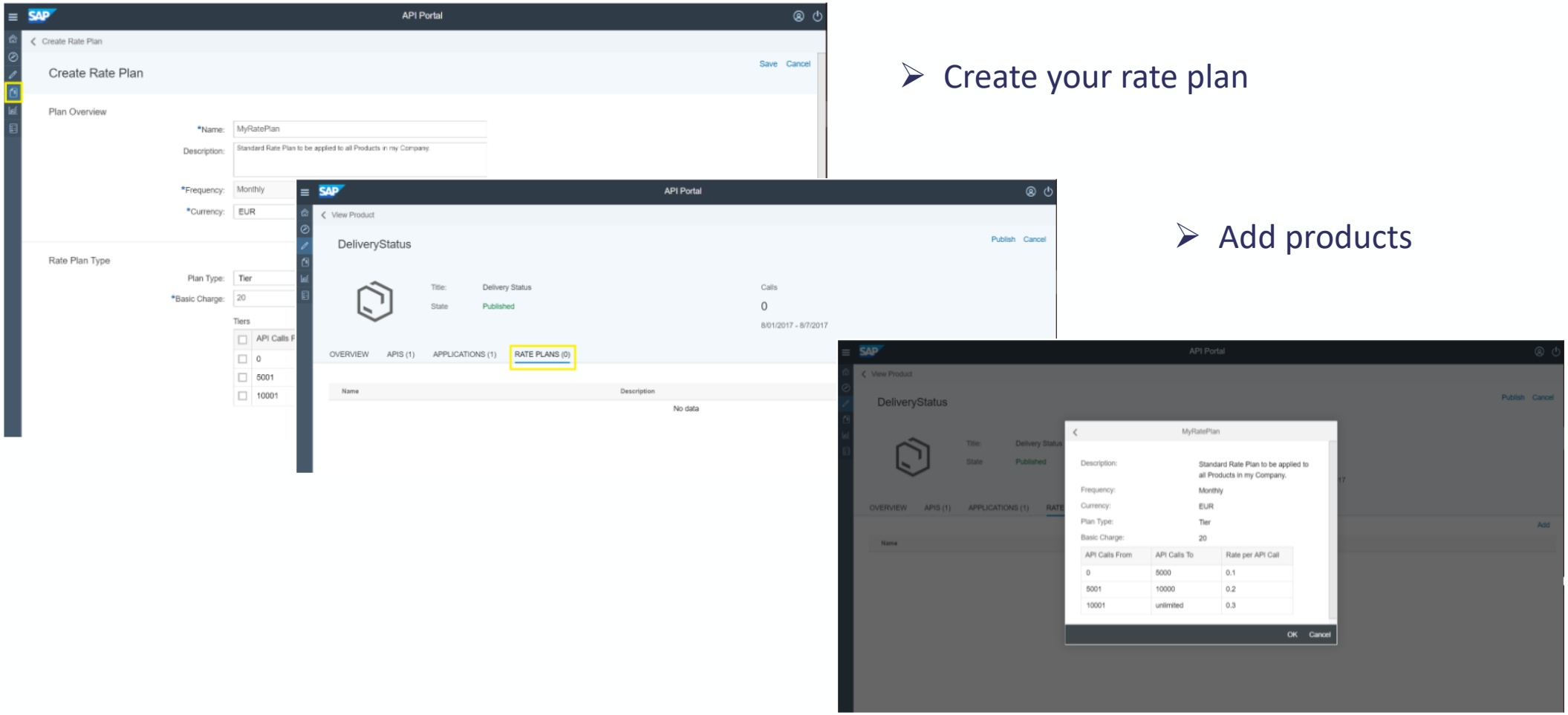
Host Alias:
s0015085037trial-trial.apim1.hanatrial.ondemand.com

API Base Path:
/s4hanacloud/sap/opu/odata/sap/API_SALES_ORDER_SRV

Description:

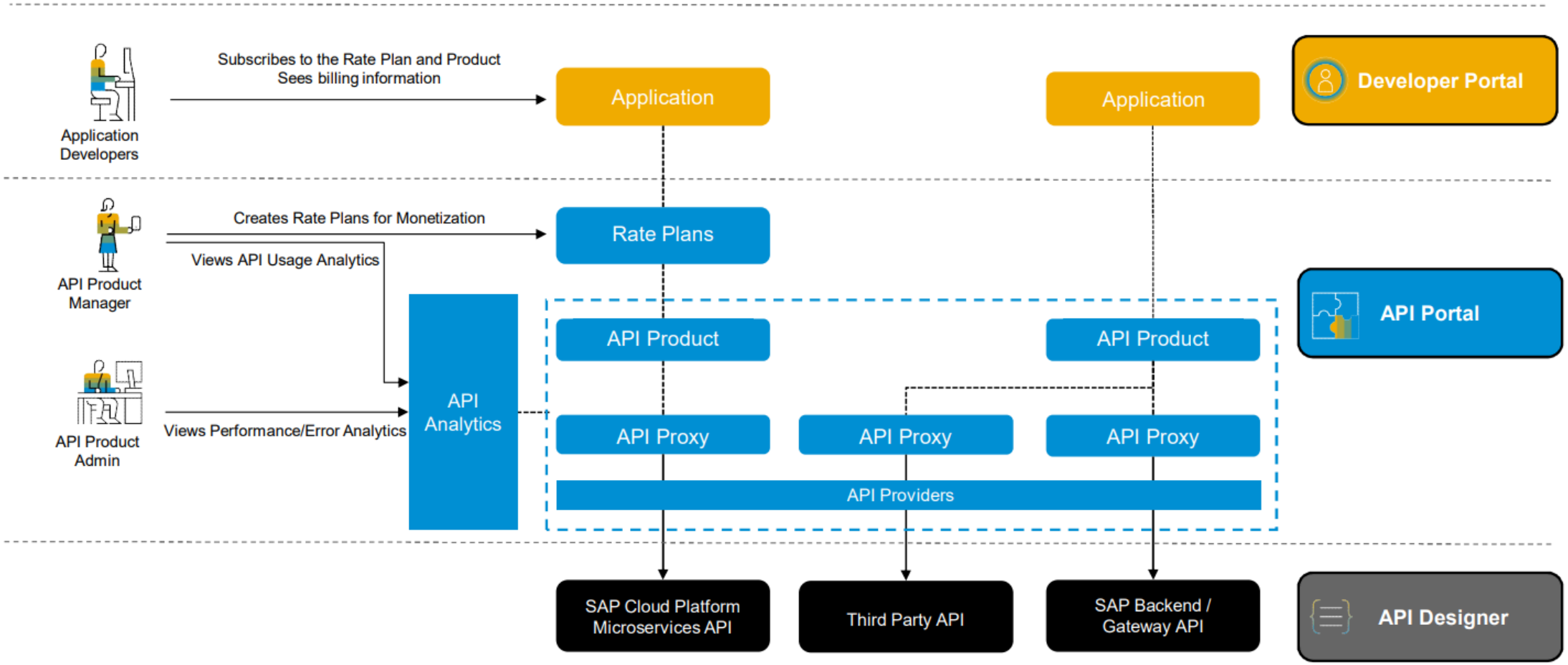
In every API call, you can make use of the following operations: - You can retrieve sales orders. Apply any of the filters provided or retrieve all existing data. - You can create sales orders. Note that you must use “deep insert” requests (a header plus the following entities: header partner, header pricing element, header text, payment details, item, item partner, item pricing element, and item text). You cannot create entities without including any related entities. - For existing sales orders, you can create new items. Note that you must use “deep insert” requests (with the following entities: item partner, item pricing element, and item text). You cannot create entities without including any related entities. - For existing sales orders, you can update the header, header partner, header pricing element, header text, payment details, item, item partner, item pricing element, and item text. - For existing sales orders, you can delete the header, header partner, header pricing element, header text, payment details, item, item partner, item pricing element, and item text. - You can accept or deny approval requests for sales orders that cannot be processed without the consent of an approver.

Monetize your API's



- Create your rate plan
- Add products

Monetize - Architecture

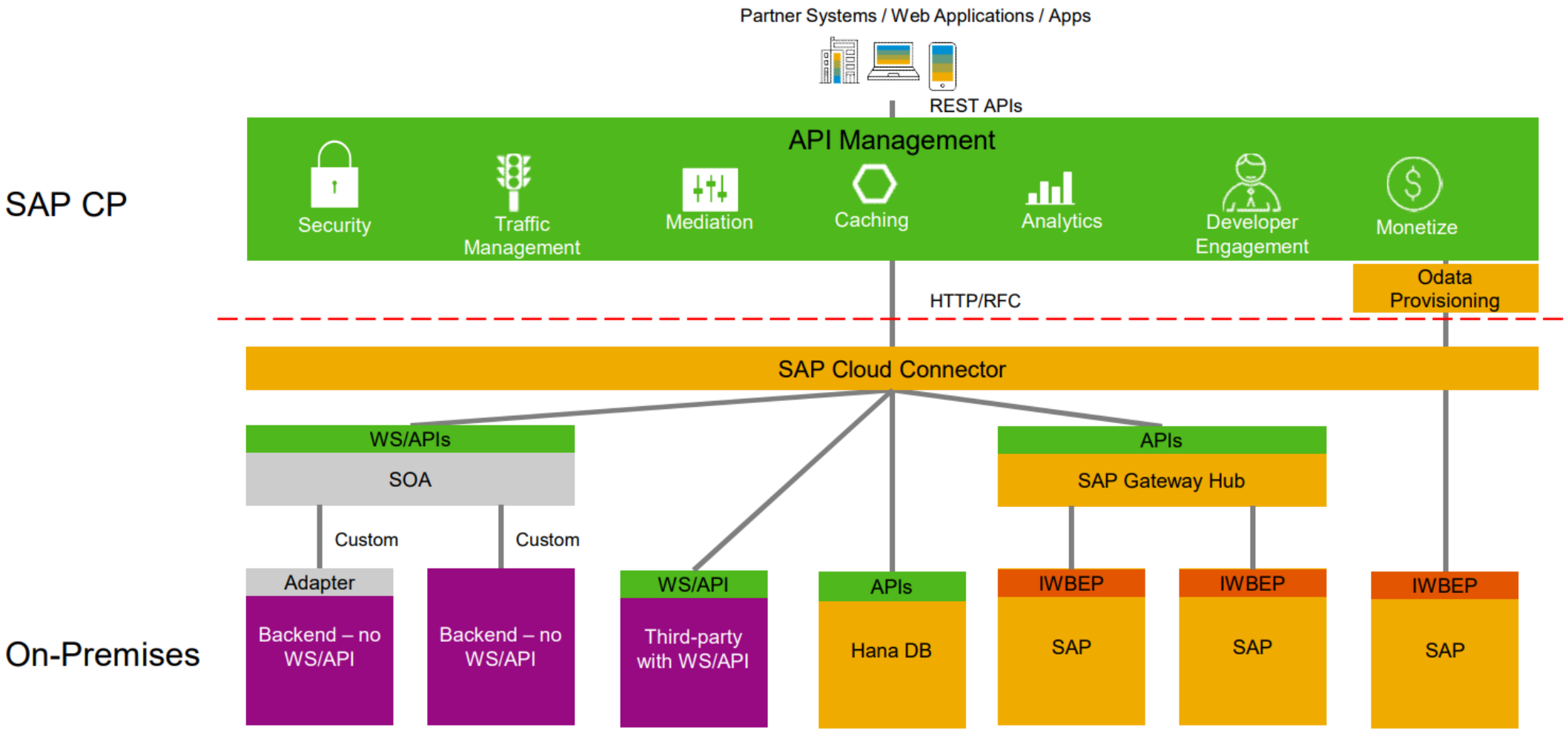


Key Features

- **Configuration instead of coding**
  - Managing APIs consists of applying standard features in the areas of security, traffic management and mediation.
  - Pre-defined within SAP API Management & need to be adapted to your needs.
  - Centrally simplify and streamline the management of APIs
- **Efficient API Publishing**
  - Simplify API consumption for the Application Developer.
  - Self-service and integrated Developer Portal
    - featuring self-service capabilities, searching, documentation, testing and subscription capabilities.
  - Reduce the time and effort of onboarding Developers, allowing for better time-to-market and reduced costs.
- **Broad analytics**
  - API usage is not limited to internal consumption
  - Monitoring and analytics are even more relevant.
  - Provides central, technical and business-driven analytic capabilities.
  - Quickly solve issues for increased user satisfaction and to steer your API program for an optimal return on investment

Architecture examples

High-Level Architecture in an SAP Ecosystem



Usefull links

- https://developers.sap.com/mission.apim-fiori-app.html
- https://blogs.sap.com/2019/04/25/sap-api-management-a-full-overview/
- https://blogs.sap.com/2016/03/03/sap-api-management-overview-getting-started/
- https://blogs.sap.com/2019/02/11/sap-api-management-engage-with-your-developer-community-on-the-brand-new-developer-portal/https://blogs.sap.com/2019/02/11/sap-api-management-engage-with-your-developer-community-on-the-brand-new-developer-portal/
- https://blogs.sap.com/2017/08/22/sap-cloud-platform-api-management-api-security-best-practices/
- https://help.sap.com/doc/980f5f55402145b1ad140359a7257229/CLOUD/en-US/Unit%202%20-%20Overview%20of%20SAP%20API%20Management%20and%20its%20Components.pdf
- Try out on SAP Cloud Platform Trial:
  - **https://blogs.sap.com/2016/02/10/how-to-use-sap-api-management-on-hcp-trial/**
- Auto-generate api documentation from Swagger Inspector for Rest API from SAP CPI
  - **https://blogs.sap.com/2019/06/19/10-steps-to-auto-generate-api-documentation-from-swagger-inspector-for-rest-apis-from-sap-cpi/**

Questions?